

Text as Data

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WT2026

Overview

- 1 From Documents to Data
- 2 Approaches to Text Analysis
- 3 Classification
- 4 Topic Modeling

Motivating Example: Political Violence & Party Formation



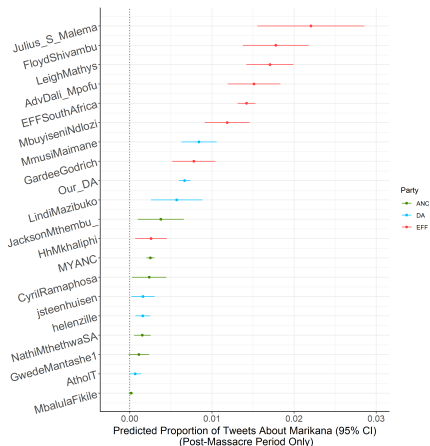
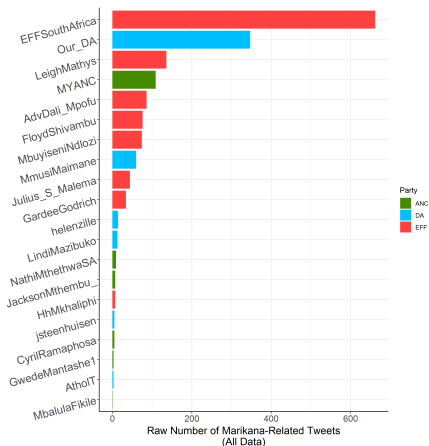
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Motivating Example: Political Violence & Party Formation



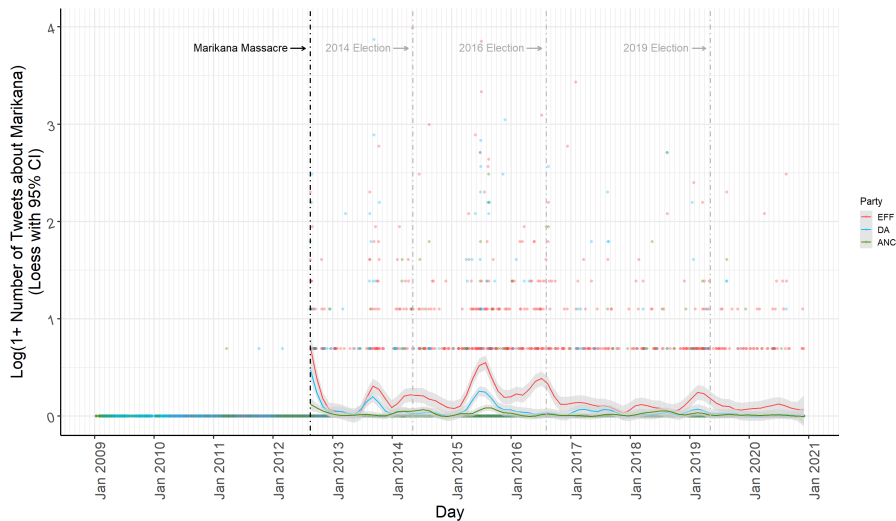
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Motivating Example: Political Violence & Party Formation

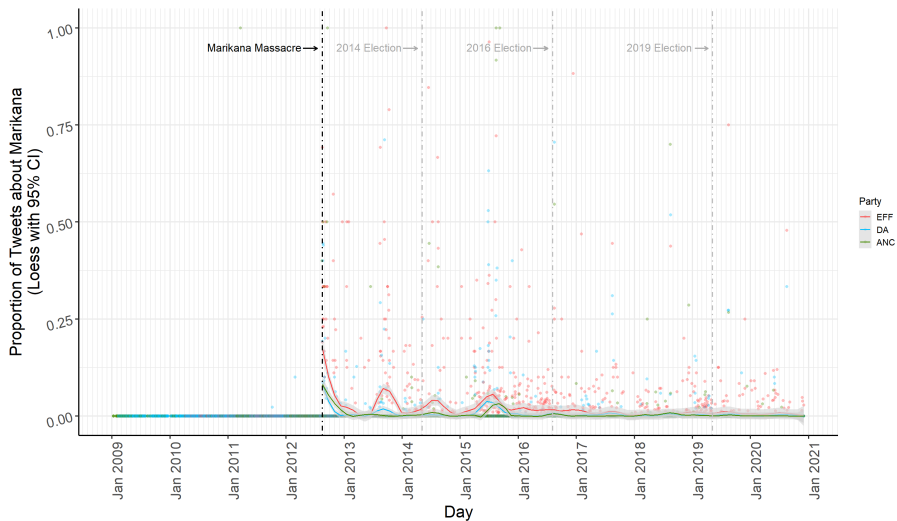


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Motivating Example: Political Violence & Party Formation



Motivating Example: Political Violence & Party Formation



Text as Data: an “embarrassment of riches”

For most of its history, text analysis in political science was **qualitative**.

Quantitative text analysis (“distant reading”) is different:

- Not focused on interpretation in light of norms or belief systems
- **Reliable and replicable**: can cope with large volumes of material
- Still requires qualitative judgements: for validation and to make sense of findings

A huge amount of politics happens through **language**:

- Political speeches, legislative debates, manifestos, lobbying documents, campaign websites
- Bills, laws, treaties, constitutions, court opinions and judgments
- News articles, opinion columns; text-based correspondence
- Open-ended survey questions, interviews, focus groups
- Twitter, Facebook, Reddit, blogs, other social media

What Can We Do with Text Data?

These are (often) **big data**: scale → **automated text analysis**

- **Classify** documents as belonging to particular types
- **Describe** the content and sentiment of text
- Use text to **inform our coding** of variables
- **Predict** outcomes based on text
- **Summarize** large amounts of text
- ... and more!

Note: This is an active area of research → new methods all the time. Different questions and datasets imply different methods. Cannot possibly cover everything in one week.

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From Documents to Data

The data begin as a structured **document** (a tweet, a bill, a speech — any unit of text).



Economic Freedom Fighters ✓
@EFFSouthAfrica



#EFFAfricaDay DSG @hlengiwe44maxon wishing all Africans an Africa Day that celebrates the resilience and tenacity of African women. #AfricaDay

Multiple documents make a **corpus** (the population of texts to be analyzed):

```
head(data)
  usernameTweet
107531 EFFSouthAfrica EFFAfricaDay DSG hlengiwe44maxon wishing all Africans an Africa Day that celebrates the resilience and tenacity of African women Africaday pic com 0sgg1ugeFS
107532 EFFSouthAfrica Africaday the indisputable ndiyagodola calls us all to hold hands and unite Africaunite pic com 1GtHxnhuqy
107533 EFFSouthAfrica Our SDG hlengiwe44maxon on some Pantsula Dance Africaday pic com RL1jTs5AMS
107534 EFFSouthAfrica Fighter ndiyagodola caught up in the spirit of Africanunity pic com P2Nf-dwv89P
107535 EFFSouthAfrica Africaday Alaska is here with us Africaunite pic com 1ouddgcJoik
107536 EFFSouthAfrica Africa Day 2018 c31X5eebNMk a via YouTube
```

Naively, this is just a dataset with a character column of “text”

The Corpus and Term-Document Matrix

In statistical software, designate the text column as a **corpus**:

```
data_text ~ Corpus(vectorSource(data$text[data$date-dates[1] & data$date<dates[2]]))
> data_text
<<simplecorpus>>
Metadata: corpus specific: 1, document level (indexed): 0
Content: documents: 245
> names(data_text[[1]])
[1] "content" "meta"
> data_text[[1]]$content
[1] "KT corp must be known that to get in bed with SA gov or work with parastatals one must pay good kickbacks telkom"
> data_text[[1]]$meta
author      : character(0)
datetimestamp: 2019-12-03 18:43:54
description : character(0)
heading     : character(0)
id          : 1
language    : en
origin      : character(0)
```

Then convert into a (cleaned, sparse) **term-document matrix**:

[illegible]

Think of as a **bag of words/n-grams** – mostly ignore structure.

Pre-processing Text

Before building the TDM, we **pre-process** the raw text:

- **Tokenize**: split text into individual words (tokens)
- **Remove stopwords**: function words (“the”, “an”, “for”) that carry no substantive content — but choices are context-dependent
- **Lowercase & strip**: remove capitalization, punctuation, HTML tags, control characters
- **Stem or lemmatize**: reduce words to root form (“running” → “run”; “saw” → “see”)
- **Minimum word length**: discard very short tokens (default: ≥ 3 characters)

Result: a **sparse matrix** — most words appear in very few documents.

Retaining some word order: **bigrams** (“White House”, “not good”) or **trigrams**. For most common tasks (sentiment, topic modeling), unigrams perform just as well.

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Generations of Text Analysis

- **Symbolic/rules-based methods**

- 1950s–2000s: hand-crafted rules and dictionaries

- **Bag of words** (e.g. **term-document matrix**)

- 1990s–2010s: count-based representations

- **Embeddings** (e.g. **Word2Vec**)

- 2013–2018: dense vector representations; captures semantic meaning

- **Transformers** (e.g. **BERT, GPT**)

- 2018–present: context-sensitive, state-of-the-art

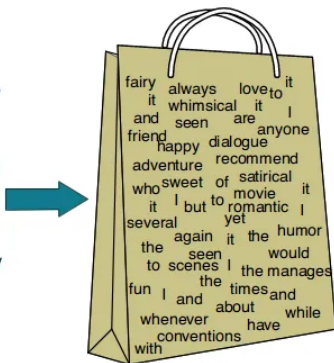
- Retrieval-Augmented Generation (RAG)?

Bag of Words

Bag of words (e.g. **term-document matrix**)

- Simple, but effective
- Good for many tasks
- Easy to understand and validate

I love this movie! It's sweet,
but with satirical humor. The
dialogue is great and the
adventure scenes are fun...
It manages to be whimsical
and romantic while laughing
at the conventions of the
fairy tale genre. I would
recommend it to just about
anyone. I've seen it several
times, and I'm always happy
to see it again whenever I
have a friend who hasn't
seen it yet!

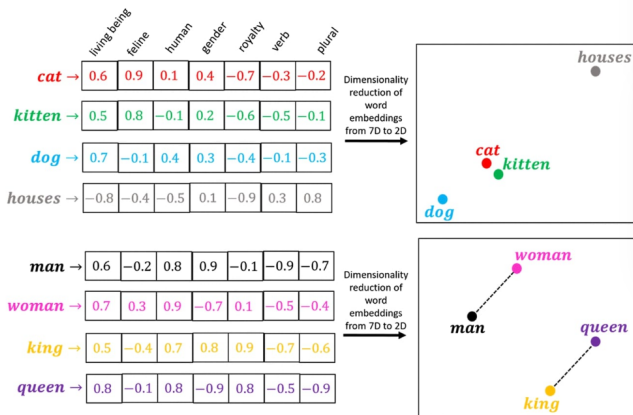


it	6
I	5
the	4
to	3
and	3
seen	2
yet	1
would	1
whimsical	1
times	1
sweet	1
satirical	1
adventure	1
genre	1
fairy	1
humor	1
have	1
great	1
...	...

Word Embeddings

Moving beyond bag of words: **word embeddings**

- Code words as **vectors** in a high-dimensional space
- Captures **semantic meaning**: similar words are close together
- Can perform arithmetic: $king - man + woman \approx queen$
- Examples: **Word2Vec**, GloVe, FastText



Embeddings Example: Word2Vec (Mikolov et al. 2013)

- Words used in similar contexts will have similar vectors
- Learns from context: “You shall know a word by the company it keeps”
- Enables arithmetic on meaning: captures relationships between concepts
- Political science application (Rodriguez & Spirling 2022): if the distance between “immigrants” and “hardworking” is smaller for liberals than conservatives, we learn something about their relative worldviews

Attention Is All You Need

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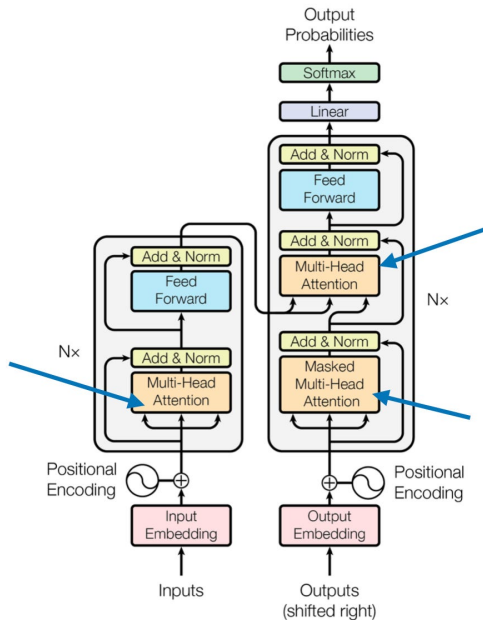
Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train. Our model achieves 28.4 BLEU on the WMT 2014 English-to-German translation task, improving over the existing best results, including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation task, our model establishes a new single-model state-of-the-art BLEU score of 41.0 after training for 3.5 days on eight GPUs, a small fraction of the training costs of the best models from the literature.

Transformers

- Word embeddings (e.g., Word2Vec) represent words as vectors but still treat text largely independently of sequence structure
- Earlier neural models (e.g., RNNs) tried to model **word order**, but processed text **sequentially**
- Transformers instead use **attention** to process sequences **in parallel**
- Produce **context-sensitive representations**: each word interpreted using surrounding words
- Two common architectures:
 - **Bidirectional** models (e.g., **BERT**) read text using left *and* right context
 - **Autoregressive** models (e.g., **GPT**) generate text one token at a time
- GPT models underlie ChatGPT and other LLMs
- **Computational advantage**: parallel architecture enables massive GPU training on huge text corpora

Transformer Architecture



Attention Mechanism in Transformers

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

The FBI is chasing a criminal on the run .

“Masked self-attention” in decoder-style transformers (GPT)

How do these representations connect to what you can do with text?

- **Machine learning** gives us two broad modes:
 - **Supervised learning**: we have labeled data; want to classify or predict
 - **Unsupervised learning**: no labels; want to discover structure
- These approaches apply across generations:
 - Bag-of-words + supervised → **dictionary methods, naive Bayes**
 - Transformers + supervised → **fine-tuned BERT**
 - Dimensionality reduction → **topic modeling (LDA, STM)**

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We now have an **enumeration** of:

- all words or n-grams (the **terms**)
- whether they are present in each corpus entry (the **documents**)

For **classification** (assigning texts to a defined set of categories):

- 1 **Dictionary methods**
- 2 **Supervised learning**

Supervised learning basic steps:

- 1 Construct a training set
- 2 Apply supervised learning method
- 3 Validate

Dictionary Methods vs. Supervised Learning

Dictionary (lexicon-based) methods: key-word frequencies signal category membership

- **Advantages:** low cost, easily replicated, applicable out-of-sample
- **Limitation:** word meaning is contextual; a word negative in one setting may be positive in another

Supervised learning: human coders classify a **training set**; algorithm learns to classify the rest

- **Advantages:** domain-specific; performance is easy to quantify and validate
- Building the codebook forces conceptual precision
- **Validate** using cross-validation (held-out test sets)

Modern Classification: Transformers

Fine-tuning pre-trained transformers (BERT, RoBERTa) for classification:

- Pre-trained on massive corpora; **fine-tuned** on your labeled examples
- Context-sensitive: “right-wing party” $\neq$ “the right answer”
- Outperforms bag-of-words on most benchmarks; works with limited labeled data

LLMs as annotators: prompt GPT-4/Claude to label political text:

- Gilardi, Alizadeh & Kubli (2023): ChatGPT outperforms crowdworkers on annotation tasks
- Fast, cheap, scalable — but risks: hallucination, prompt sensitivity, opacity

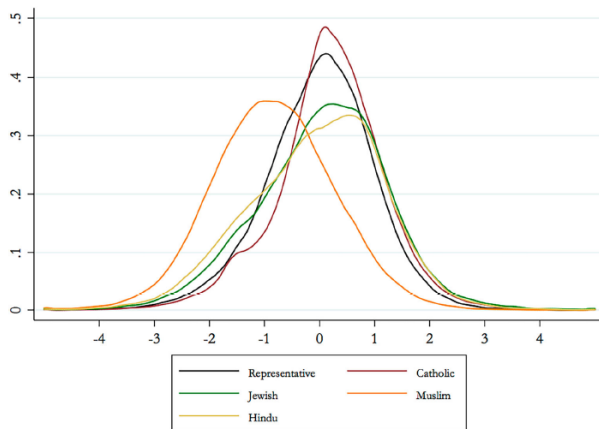
Applied examples:

- Bill classification, fine-tuned BERT — Ornstein et al. (2022)
- Cross-lingual populism, 26 countries — Müller & Proksch (2024)

Example: Sentiment Analysis (Bleich & van der Veen 2021)

- Authors ask **whether Muslims are portrayed more negatively** than other religious groups in the media, and under what conditions
- Collect 850,000 articles mentioning Muslims, Hindus, Jews, or Catholics in 17 US newspapers, 1996–2015; compare to a representative baseline
- **Approach**: dictionary (lexicon-based) sentiment analysis; average scores across 8 lexica
- **Covariates**: US vs. non-US events, post-9/11, partisan lean of the newspaper
- **Main finding**: articles about Muslims are more negative than baseline and more negative than other groups

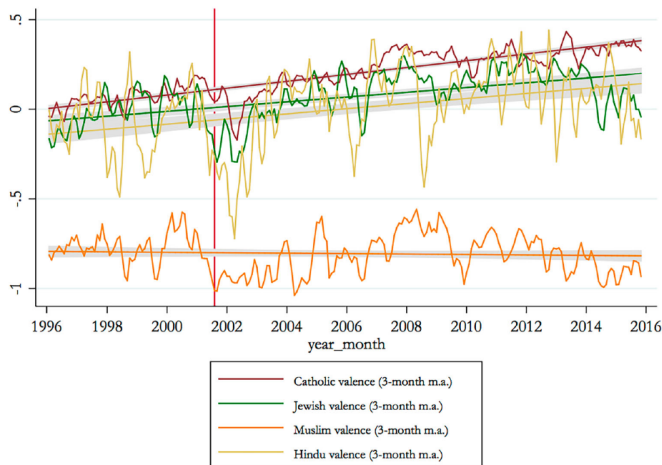
Sentiment Analysis (Bleich & van der Veen 2021)



Valence distribution estimates (kernel density) of articles by religion compared to the representative corpus.

Bleich, E. and van der Veen, A.M., 2021. Media portrayals of Muslims: a comparative sentiment analysis of American newspapers, 1996–2015. *Politics, Groups, and Identities*, 9(1), pp.20–39.

Sentiment Analysis (Bleich & van der Veen 2021)



Three-month moving average valences of articles mentioning religious groups.

Bleich, E. and van der Veen, A.M., 2021. Media portrayals of Muslims: a comparative sentiment analysis of American newspapers, 1996–2015. *Politics, Groups, and Identities*, 9(1), pp.20–39.

Sentiment Analysis (Bleich & van der Veen 2021)

Table 3. Sample preprocessed sentences at standard deviation valence intervals.

Valence score	Sample preprocessed sentences
-2	• That has been followed by the taking over of most of the country by the Taliban a fundamentalist Islamic movement that considers much of western culture to be evil • A Muslim man charged with setting fire to a Marietta mosque may be in the country illegally law enforcement officials confirmed Thursday
-1	• Muslims have been silent and even maybe passive members of the society Rehana Jan said • Separately Bosnian Muslims rallied yesterday to demand to return to a Serb-held town
0	• As a foreign correspondent I spent 13 of 17 years close to Muslims as a colleague and friend first at the UN then in India, Pakistan and Afghanistan • I am talking about the major world religions Judaism, Islam, and Christianity
1	• Colleagues said they expected her to ask secretary of state Colin I Powell one of the administration's most popular figures to embark on a listening tour in crucial Muslim nations • The journey is not a required part of the annual pilgrimage or hajj but many Muslims take the chance to visit Islam's second most holy site
2	• Arizona is home to infinite spiritual expressions from the centuries-old Catholic tradition brought by the early Jesuit and Franciscan priests to the Muslims who call modern-day Arizona home • Each food kit was enriched with extra vitamins and designed to be acceptable to the Muslim diet while also supplying 2300 calories enough nutrition for about a day

Bleich, E. and van der Veen, A.M., 2021. Media portrayals of Muslims: a comparative sentiment analysis of American newspapers, 1996–2015. *Politics, Groups, and Identities*, 9(1), pp.20–39.

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Topic Modeling (Unsupervised Learning)

Latent Dirichlet Allocation (LDA) (aka topic modeling):

- Method of **unsupervised** classification
- Number of topics predetermined by the researcher
- “Bag of words” assumption; data are pre-processed
- A document can belong to multiple topics
- Easy to implement: `topicmodels` and `tm` packages in R

Structural Topic Modeling (STM) (Roberts, Stewart, et al. 2014):

- A popular **refinement on LDA** (there are many others)
- Allows for **metadata** (document-level covariates)
- Allows for direct **hypothesis testing**
- Easy to implement: `stm` package in R

Structural Topic Models for Open-Ended Survey Responses

Margaret E. Roberts University of California, San Diego
Brandon M. Stewart Harvard University
Dustin Tingley Harvard University
Christopher Lucas Harvard University
Jetson Leder-Luis California Institute of Technology
Shana Kushner Gadarian Syracuse University
Bethany Albertson University of Texas at Austin
David G. Rand Yale University

Collection and especially analysis of open-ended survey responses are relatively rare in the discipline and when conducted are almost exclusively done through human coding. We present an alternative, semiautomated approach, the structural topic model (STM) (Roberts, Stewart, and Airolidi 2013; Roberts et al. 2013), that draws on recent developments in machine learning based analysis of textual data. A crucial contribution of the method is that it incorporates information about the document, such as the author's gender, political affiliation, and treatment assignment (if an experimental study). This article focuses on how the STM is helpful for survey researchers and experimentalists. The STM makes analyzing open-ended responses easier, more revealing, and capable of being used to estimate treatment effects. We illustrate these innovations with analysis of text from surveys and experiments.

Beyond LDA: Embedding-Based Topic Models

New approaches use **pre-trained embeddings**:

- **BERTopic** (Grootendorst 2022):

1. Use BERT to convert each document into a vector that captures its semantic meaning
 2. Reduce dimensions (UMAP)
 3. Cluster documents (HDBSCAN)
 4. Extract topic keywords
- minimal pre-processing required

- **Contextualized Topic Models (CTM)** (Bianchi et al. 2021):

- Combines BERT sentence embeddings with a neural topic model

- **Top2Vec**: topics as dense regions in document embedding space

Advantages over LDA/STM:

- Handles proper nouns, multi-word phrases, abbreviations better than BoW
- Semantic similarity in embedding space helps produce more coherent topics
- Multilingual: use multilingual BERT (mBERT) for cross-national corpora

Topic Modeling in Politics Research: Applications

- **Senators' styles**: constituency communication in press releases (Grimmer 2010, 2013)
- **Human rights**: freedom of expression claims, human rights recommendations and monitoring in UN reports (Bagozzi & Berliner 2018)
- **Radical rhetoric**: ideological styles in militant texts (Karell & Freedman 2019)
- **Agenda dynamics**: topic shifts in legislative deliberation based on parliamentary speeches and deliberation (Rossiter 2021)

Newer methods (embeddings / LLMs):

- **Emotive rhetoric**: word embeddings to score emotional language and affective polarization in speeches (Osnabrügge, Hobolt & Rodon 2021)
- **Cross-lingual political rhetoric**: study prevalence of nostalgia in party manifestos of across 24 European democracies since 1946 (Müller & Proksch 2024)